

Exercise for chapter 8

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Chapter 8

8.1. Demonstrate the law of corresponding states by writing the van der Waals equation in terms of the reduced variables. Calculate the compressibility factor for a van der Waals fluid at its critical point and compare the result with the values obtained for real gases at their critical points listed in Table 8.1. Calculate the value of $(\partial U/\partial V)_T$ for a van der Waals fluid.

Chapter 8

Solution to 8.1

For the mole

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

$$\left(P_R P_{cr} + \frac{a}{V_R^2 V_{cr}^2}\right)(V_R V_{cr} - b) = RT_R T_{cr}$$

$$\left(P_R \frac{R}{27b^2} + \frac{a}{V_R^2 9b^2}\right)(V_R 3b - b) = RT_R \frac{8a}{27bR}$$

$$\frac{ab}{27b^2} \left(P_R + \frac{3}{V_R^2}\right)(3V_R - 1) = \frac{8a}{27b} T_R$$

$$\left(P_R + \frac{3}{V_R^2}\right)(3V_R - 1) = 8T_R$$

Chapter 8

Solution to 8.1

$$\frac{P_{cr} V_{cr}}{RT_{cr}} = \frac{a}{27b^2} \frac{3b}{R} \frac{27bR}{8a} = \frac{3}{8} = 0.375$$

$$dU = TdS - PdV$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial r}\right)_T - P = T\left(\frac{\partial P}{\partial T}\right)_r - P$$

For a van der Waals gas

$$P = \frac{RT}{V - b} - \frac{a}{V^2}$$

So $\left(\frac{\partial P}{\partial T}\right)_r = \frac{R}{V - b}$ and $\left(\frac{\partial V}{\partial r}\right)_T = \frac{RT}{V - b} - P = \frac{a}{V^2}$

Chapter 8

8.2 n moles of an ideal gas A and $(1 - n)$ moles of an ideal gas B, each at 1 atm pressure, are mixed at total constant pressure. What ratio of A to B in the mixture maximizes the decrease in the Gibbs free energy of the system? If the decrease in the Gibbs free energy is ΔG^M , to what value must the pressure be increased in order to increase the Gibbs free energy of the gas mixture by $1/2 \Delta G^M$?

Chapter 8

Solution to 8.2

Mix n moles of A and $(1-n)$ moles of B at constant $P=1$ atm

$$\Delta G^{Mix} = RT[n \ln n + (1 - n) \ln(1 - n)]$$

$$\begin{aligned} \frac{d\Delta G^{Mix}}{dn} &= RT[\ln n + 1 - \ln(1 - n) - 1] = RT \ln \frac{n}{1 - n} \\ &= 0 \quad \text{when } n = 0.5 \end{aligned}$$

Chapter 8

Solution to 8.2

So Maximum decrease occurs when $n_A = n_B = 0.5$

Increasing the pressure from 1 atm to P atm

increases the free energy by

$$\Delta G = RT \ln P$$

So
$$-\frac{1}{2} \Delta G^{Mix} = RT \ln P = \frac{1}{2} RT \ln 2$$

So
$$P = \sqrt{2} = 1.414 \text{ atm}$$

Chapter 8

8.3 You are responsible for the purchase of oxygen gas which, before use, will be stored at a pressure of 200 atm at 300 K in a cylindrical vessel of diameter 0.2 m and height 2 m. Would you prefer that the gas behaved ideally or as a van der Waals fluid? The van der Waals constants for oxygen are $a = 1.36 \text{ liters}^2 \text{ atm mole}^{-2}$ and $b = 0.0318 \text{ liters/mole}$.

Chapter 8

Solution to 8.3

The volume of the tank is

$$\pi r^2 h = 3.142 \times 0.1^2 \times 2 = 0.06284 m^3 = 62.84 l$$

Thus if the gas is ideal, the tank contains

$$n = \frac{PV'}{RT} = \frac{200 \times 62.84}{0.08206 \times 300} = 511 moles$$

If the gas is a van der Waals gas

$$\left(P + \frac{an^2}{V^2} \right) (V' - nb) = nRT$$

Chapter 8

Solution to 8.3

$$\left(200 + \frac{1.36n^2}{62.84^2}\right)(62.84 - 0.0318n) = n \times 0.08206 \times 300$$

Which gives $n=565$ moles

It is thus less expensive to fill the tank if the gas behavior is ideal

Chapter 8

8.4 The virial equation of state for n-butane at 460 K is $Z = 1 + A/V + B/V^2$, in which $A = -265 \text{ cm}^3/\text{g} \cdot \text{mole}$ and $B = 30250 \text{ cm}^6/\text{g} \cdot \text{mole}^2$. Calculate the work required to reversibly compress 1 mole of n-butane from 50 to 100 atm at 460 K.

Chapter 8

Solution to 8.4

$$Z = 1 - \frac{265}{V} + \frac{30250}{V^2}$$

So

$$P(atm) = 82.057 \times 460 \left[\frac{1}{V} - \frac{265}{V^2} + \frac{30250}{V^3} \right]$$

At $P_1 = 50 atm$, this give $V_1 = 394.6 cm^3$

And at $P_2 = 100 atm$, $V_2 = 176.9 cm^3$

Chapter 8

Solution to 8.4

So

$$w = \int_1^2 P dV = RT \int_1^2 \left[\frac{1}{V} - \frac{265}{V^2} + \frac{30250}{V^3} \right] dV$$
$$= RT \left[\ln \frac{V_2}{V_1} + 265 \left(\frac{1}{V_2} - \frac{1}{V_1} \right) - \frac{30250}{2} \left(\frac{1}{V_2^2} - \frac{1}{V_1^2} \right) \right]$$
$$= 8.3144 \times 460 \left[\ln \frac{176.9}{394.6} + 265 \left(\frac{1}{176.9} - \frac{1}{394.6} \right) - \frac{30250}{2} \left(\frac{1}{176.9^2} - \frac{1}{394.6^2} \right) \right]$$
$$= -1384J$$

Chapter 8

- 8.5** For sulfur dioxide, $T_{cr} = 430.7K$ and $P_{cr} = 77.8atm$. Calculate
- The critical van der Waals constants for the gas.
 - The critical volume of van der Waals SO_2 .
 - The pressure exerted by 1 mole of SO_2 occupying a volume of 500 cm^3 at 500 K. Compare this with the pressure which would be exerted by an ideal gas occupying the same molar volume at the same temperature

Chapter 8

Solution to 8.5

$$T_{cr} = \frac{8a}{27bR}$$

$$P_{cr} = \frac{a}{27b^2}$$

$$\frac{T_{cr}}{P_{cr}} = \frac{8b}{R}$$

So

$$b = \frac{430.7 \times 0.08206}{77.8 \times 8} = 0.0567 \text{ l/mole}$$

$$a = 27b^2P_{cr} = 27 \times 0.0567^2 \times 77.8 = 6.77 \frac{\text{atm} \cdot \text{l}^2}{\text{mole}^2}$$

Chapter 8

Solution to 8.5

$$V_{cr} = 3b = 0.170 \text{ l/mole}$$

$$P(\text{van der Waals}) = \frac{RT}{V - b} - \frac{a}{V^2} = \frac{0.08206 \times 500}{0.5 - 0.0567} - \frac{6.77}{0.5^2} \\ = 65.5 \text{ atm}$$

$$P(\text{ideal gas}) = \frac{RT}{V} = \frac{0.08206 \times 500}{0.5} = 82.1 \text{ atm}$$

Chapter 8

8.6 One hundred moles of hydrogen gas at 298 K are reversibly and isothermally compressed from 30 to 10 liters. The van der Waals constants for hydrogen are $a = 0.2461 \text{ liter}^2 \cdot \text{atm} \cdot \text{mole}^{-2}$ and $b = 0.02668 \text{ liter/mole}$, and in the range of pressure 0–1500 atm, the virial equation for hydrogen is $PV = RT(1 + 6.4 \times 10^{-4} P)$. Calculate the work that must be done on the system to effect the required change in volume and compare this with the values that would be calculated assuming that (a) hydrogen behaves as a van der Waals fluid and (b) hydrogen behaves as an ideal gas.

Chapter 8

Solution to 8.6

$$n = 100 \text{ moles}, T = 298\text{K}, V_1 = 30\text{l}, V_2 = 10\text{l}$$

$$PV = nRT(1 + 6.4 \times 10^{-4}P)$$

$$w = \int P dV$$

$$V = nRT \left(\frac{1}{P} + 6.4 \times 10^{-4} \right)$$

$$dV = nRT d \left(\frac{1}{P} \right) = -nRT \frac{dP}{P^2}$$

So

$$w = nRT \ln \frac{P_1}{P_2}$$

Chapter 8

Solution to 8.6

$$PV - 6.4 \times 10^{-4} RT P n = nRT$$

$$P = \frac{nRT}{V - n6.4 \times 10^{-4} RT}$$

So
$$P_1 = \frac{100 \times 0.08206 \times 298}{30 - 100 \times 6.4 \times 10^{-4} \times 0.08206 \times 298} = 86 \text{ atm}$$

$$P_2 = \frac{100 \times 0.08206 \times 298}{10 - 100 \times 6.4 \times 10^{-4} \times 0.08206 \times 298} = 290 \text{ atm}$$

So
$$w = 100 \times 8.3144 \times 298 \ln \frac{86}{290} = -301 \text{ kJ}$$

Chapter 8

Solution to 8.6

Van der Waals:

$$P = \frac{nRT}{V - bn} - \frac{an^2}{V^2}$$

$$a = 0.2461 \frac{\text{atm l}^2}{\text{mole}}, b = 0.02668 \text{ l/mole}$$

$$\begin{aligned} w &= \int P dV = \int \left(\frac{nRT}{V - nb} - \frac{an^2}{V^2} \right) dV \\ &= nRT \ln \left(\frac{V_2 - nb}{V_1 - nb} \right) - an^2 \left(\frac{1}{V_1} - \frac{1}{V_2} \right) \\ &= -309 \text{ kJ} \end{aligned}$$

Ideal gas

$$w = nRT \ln \frac{V_2}{V_1} = 100 \times 8.3144 \times 298 \ln \frac{10}{30} = -272 \text{ kJ}$$

Chapter 8

8.7 Using the virial equation of state for hydrogen at 298 K given in Problem 8.6, calculate

- a. The fugacity of hydrogen at 500 atm and 298 K
- b. The pressure at which the fugacity is twice the pressure
- c. The change in the Gibbs free energy caused by a compression of 1 mole of hydrogen at 298 K from 1 to 500 atm What is the magnitude of the contribution to (c) caused by the nonideality of hydrogen?

Chapter 8

Solution to 8.7

a)
$$\frac{PY}{RT} = Z = 1 + 6.4 \times 10^{-4} P$$

So
$$Z - 1 = 6.4 \times 10^{-4} P$$

$$\ln \frac{f}{P} = \int_0^P \frac{Z - 1}{P} dP = 6.4 \times 10^{-4} (500 - 0) = 0.3194$$

So
$$\frac{f}{P} = 1.376, f = 1.376 \times 500 = 688 \text{ atm}$$

Chapter 8

Solution to 8.7

b)

$$\frac{f}{P} = 2$$

So

$$\ln \frac{f}{P} = 0.693 = 6.4 \times 10^{-4}(P - 0)$$

$$\frac{f}{P} = 2 \text{ at } P = 1083 \text{ atm}$$

Chapter 8

Solution to 8.7

c)

$$\Delta G = \int V dP$$

$$V = RT \left(\frac{1}{P} + 6.4 \times 10^{-4} \right)$$

$$\begin{aligned} \Delta G &= RT \ln \frac{P_2}{P_1} + RT 6.4 \times 10^{-4} (P_2 - P_1) \\ &= 8.3144298 \ln \frac{500}{1} + 8.3144298 6.4 \times 10^{-4} (500 - 1) \\ &= 15400 + 790 = 16190J \end{aligned}$$

And the contribution due to the non-ideal behavior is 790J.

Chapter 8

8.8 Show that the truncated Kamerlingh Onnes virial equation

$$\frac{PV}{RT} = 1 + \frac{B'(T)}{T}$$

reduces to $P(V - b') = RT$, and find the expression for b'

Chapter 8

Solution to 8.8

$$\frac{PV}{\left(1 + \frac{B'(T)}{V}\right)} = RT$$

If $V \gg B'(T)$

then using the expansion

$$\frac{1}{1+x} = 1 - x \quad \text{For } x \ll 1$$

$$PV \left(1 - \frac{B'(T)}{V}\right) = P(V - B'(T)) RT$$

So $b' = B'(T)$

Chapter 8

8.9 Figure 8.8 shows an isothermal variation, with pressure, of the volume of a van der Waals fluid at a temperature below its critical temperature. Redraw this figure and below it sketch the Helmholtz free energy versus volume plot. Be sure to indicate important points on the A versus V plot.

Chapter 8

Solution to 8.9

$$dA_T = -PdV$$

$$\left(\frac{\partial A}{\partial V}\right)_T = -P$$

so the slope should always be < 0

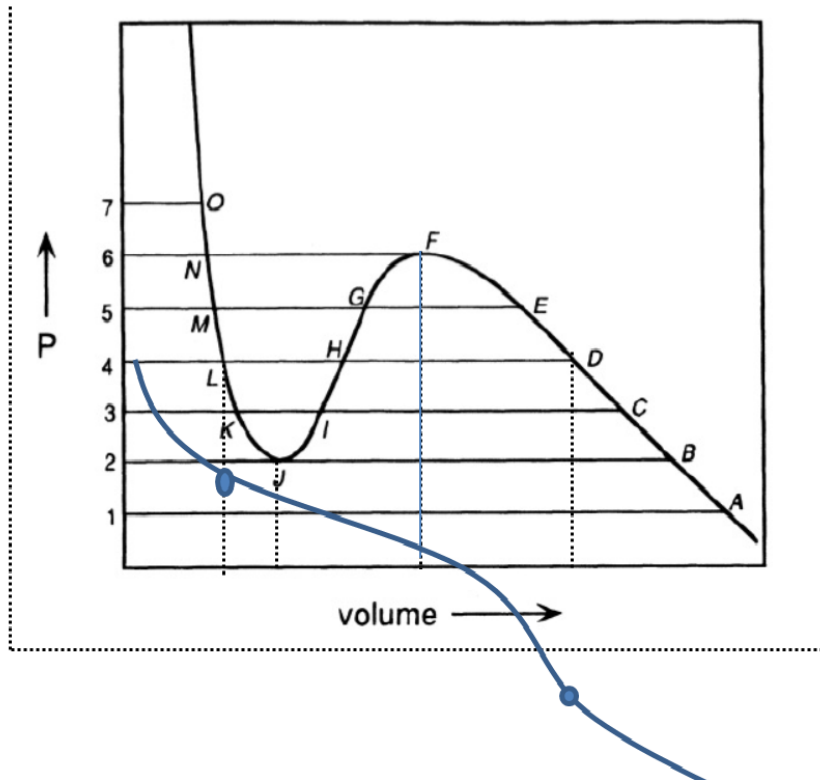
$$\left(\frac{\partial^2 A}{\partial V^2}\right)_T = \left(\frac{\partial P}{\partial V}\right)_T$$

and there are two extrema in P vs V

Chapter 8

Solution to 8.9

Thus there are two points where A vs. V changes its curvature. Also the common tangent delineates the molar volumes of the equilibrium phases.



Chapter 8

8.10 Obtain a simplified expression for the total derivative of the pressure for a van der Waals gas.

Chapter 8

Solution to 8.10

$$dP = \left(\frac{\partial P}{\partial T} \right)_V dT + \left(\frac{\partial P}{\partial V} \right)_T dV$$

$$P = \frac{RT}{(V - b)} - \frac{a}{V^2} \quad \text{for a van der Waals' gas}$$

$$\left(\frac{\partial P}{\partial T} \right)_V = \frac{R}{V - b} \quad \text{for a van der Waals' gas}$$

$$\left(\frac{\partial P}{\partial V} \right)_T = -\frac{RT}{(V - b)^2} + \frac{2a}{V^3} \quad \text{for a van der Waals' gas}$$

Chapter 8

Solution to 8.10

Thus

$$dP = \frac{R}{V-b} dT - \left(\frac{RT}{(V-b)^2} + \frac{2a}{V^3} \right) dV$$

which reduces to

$$dP = \frac{R}{V} dT - \frac{RT}{V^2} dV \quad \text{if } a = 0 \text{ and } b = 0$$

Chapter 8

8.11 Obtain a simplified expression for the total derivative of the pressure for a gas that obeys the truncated Kamerlingh Onnes virial equation:

$$\frac{PV}{RT} = 1 + \frac{B'(T)}{T}$$

Chapter 8

Solution to 8.11

$$P = \left(1 + \frac{B'T}{V}\right) \frac{RT}{V}$$

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{V} + \frac{B'(T)R}{V^2} + \frac{RT}{V^2} \left(\frac{\partial B}{\partial T}\right)$$

$$\left(\frac{\partial P}{\partial V}\right)_T = -\frac{2B'T}{V^3} - \frac{RT}{V^2}$$

$$dP = \left[\frac{R}{V} + \frac{B'(T)R}{V^2} + \frac{RT}{V^2} \left(\frac{\partial B'(T)}{\partial T}\right) \right] dT - \left[\frac{RT}{V^2} + \frac{2B'(T)RT}{V^3} \right] dV$$

which reduces to $dP = \frac{P}{V} dT - \frac{RT}{V^2} dV$ if $B'(T) = 0$

Chapter 8

8.12 Derive an expression for Z_{cr} of a van der Waals fluid

Chapter 8

Solution to 8.12

The equation of state for the van der Waals fluid is

$$P = \frac{RT}{V - b} - \frac{a}{V^2}$$

Where P is the pressure in P_a

R is the gas constant (8.314 J/K)

V is the molar volume of the gas (m^3)

a and b are empirical constants that vary from gas to gas.

To calculate Z_{cr} we must obtain: P_{cr} , V_{cr} and T_{cr}

Chapter 8

Solution to 8.12

At the critical pressure and volume, the critical isotherm has a horizontal inflection:

$$\left(\frac{\partial P}{\partial V}\right)_{T_C} = \left(\frac{\partial^2 P}{\partial V^2}\right)_{T_C} = 0$$

$$\left(\frac{\partial P}{\partial V}\right)_{T_C} = -\frac{RT}{(V-b)^2} + \frac{2a}{V^3} = 0$$

$$\left(\frac{\partial^2 P}{\partial V^2}\right)_{T_C} = \frac{2RT}{(V-b)^3} - \frac{6a}{V^4} = 0$$

Solving we get:

$$RT_{cr} = \frac{8a}{27b} \quad V_{cr} = 3b \quad P_{cr} = \frac{a}{27b^2}$$

$$Z_{cr} = \frac{P_{cr}V_{cr}}{RT_{cr}} = \frac{3}{8}$$

Chapter 8

8.13 The ideal entropy of mixing of a binary solution is given as

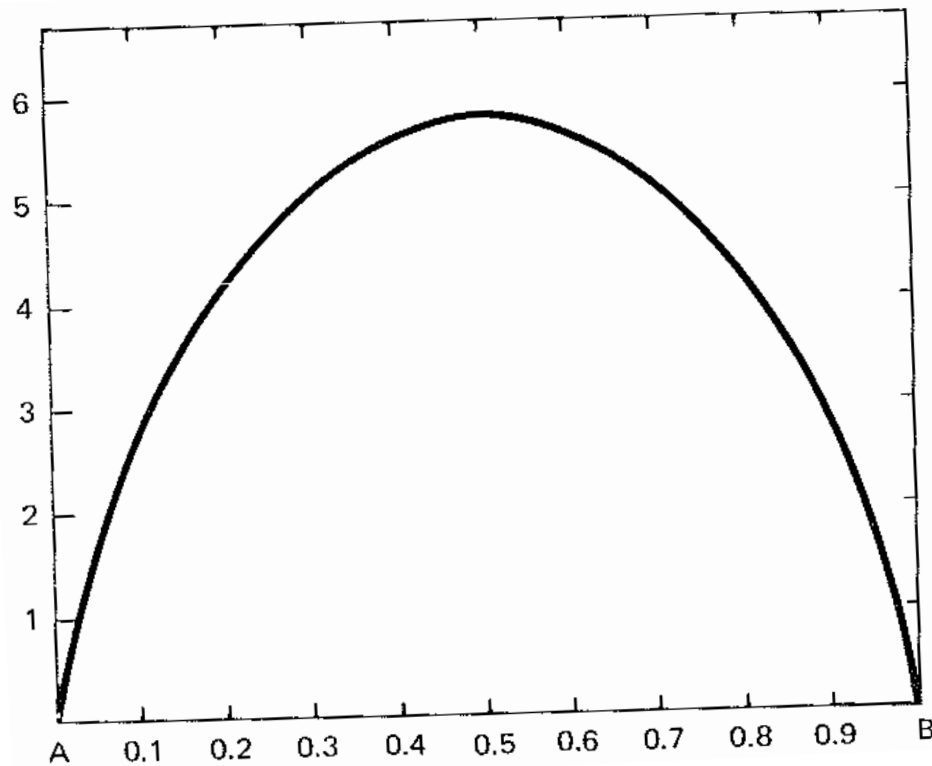
$$\Delta S_{mix} = -R(X_A \ln X_A + X_B \ln X_B)$$

- Sketch the entropy of mixing versus X_B .
- Calculate (using the expression for the ideal entropy of mixing) the ideal entropy of mixing at $X_B = 0.25$.
- Calculate the slope of this curve at $X_B = 0.25$.
- Calculate the partial molar entropy of component B for the composition $X_B = 0.25$.
- Calculate the partial molar entropy of component A for the composition $X_B = 0.25$.
- Show that your answers in (d) and (e) are consistent with your answer in (c).

Chapter 8

Solution to 8.13

a)



Chapter 8

Solution to 8.13

$$\text{b)} \quad \Delta S_{mix} = -R(X_A \ln X_A + X_B \ln X_B) = 4.56 \text{ J/mol}$$

$$\text{c)} \quad \Delta S_{mix} = -R(X_A \ln X_A + X_B \ln X_B)$$

$$\frac{d\Delta S_{mix}}{dX_B} = -R[\ln X_B - \ln X_A] = R \ln 3 = 9.13 \text{ J/mol}$$

$$\text{d)} \quad \overline{S}_B = -R \ln X_B = R \ln 4 = 11.52 \text{ J/mol}$$

$$\text{e)} \quad \overline{S}_A = -R \ln X_A = R \ln \frac{3}{4} = 2.39 \text{ J/mol}$$

$$\text{f)} \quad \frac{d\Delta S_{mix}}{dX_B} = R \ln 3 = 9.13 \text{ J/mol}$$

$$\Delta S_{mix} = \overline{S}_B - \overline{S}_A = R \ln 3 = 9.13 \text{ J/mol}$$

Thermodynamics of Materials & Phase transformations

The End

